

TEAM FOUR CHADS

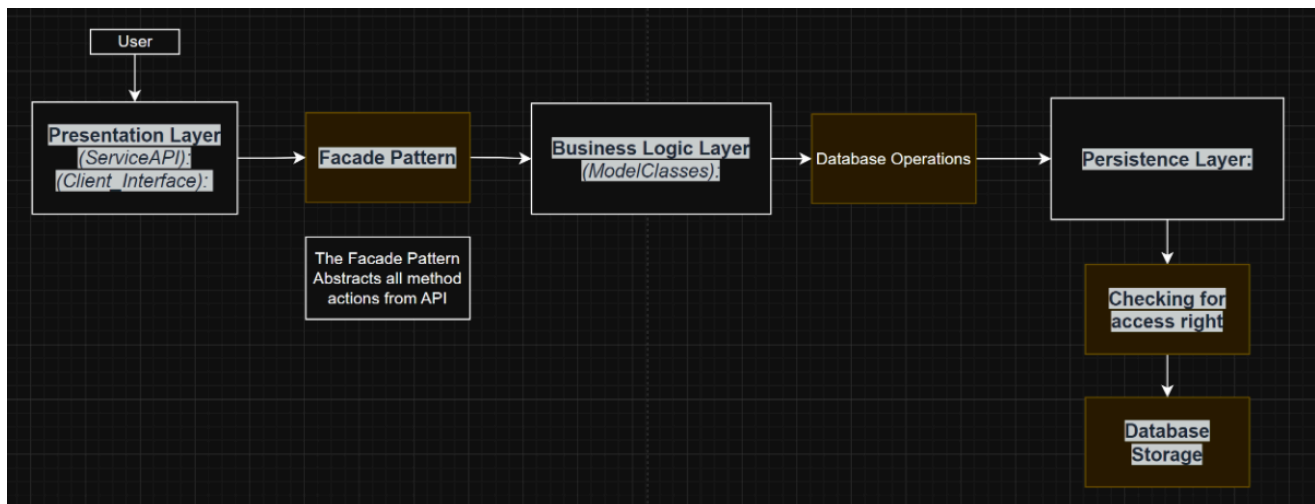
PROJECT HBNB

This project is related to developing a HBNB (Holberton Bed and Breakfast).
The Technical document you are reading explains the first structured concept in developing the framework for this project.

We have created diagrams that define the groundwork for our project:

1. High Level Package Diagram
2. Class Diagram
3. Sequence Diagram

1. High Level Package Diagram



The high level package diagram has been divided to represent three layers:

Presentation Layer

This is the UI layer which displays the website to the user and is responsible for handling the interactions between users and the HBNB system. The presentation layer, being the front end focuses on making it easier for users to communicate with the backend and database through easy to use interfaces.

Layer with most basic methods

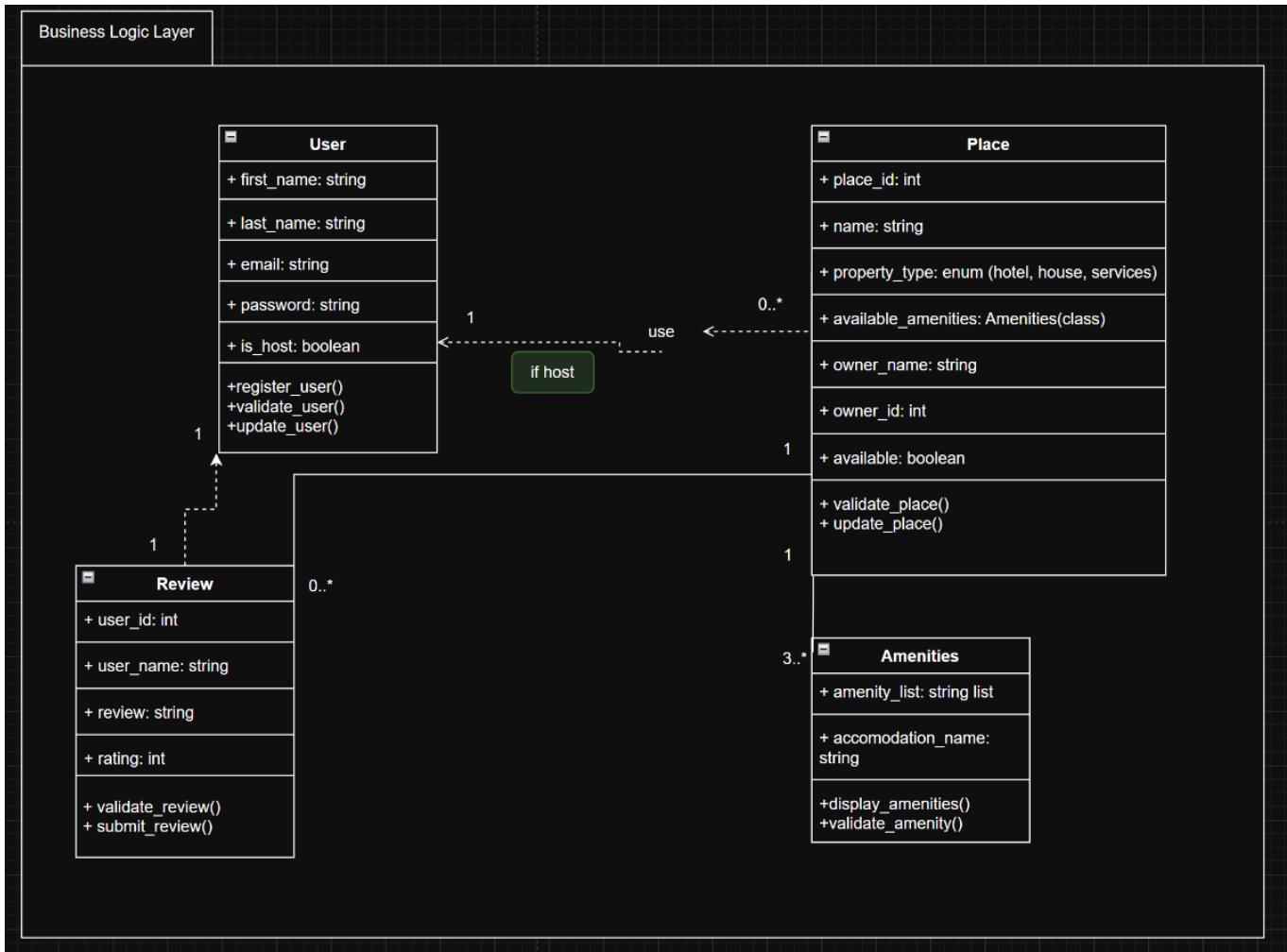
Business Logic Layer

This is the layer which contains the bulk of the website's functionality and logic. Whatever action taken by the user in the presentation layer is being acted on in the business logic layer. This layer contains the majority of our classes that has been illustrated in the upcoming class diagram.

Persistence Layer:

This layer is the database layer, where storage of information and data retrieval occurs, protected by the business logic layer. It also will be the layer that provides simple operations such as saving, updating, deleting and retrieving of objects to and from the Business layer and will hide its internal logic from the Business layer, but will be able to communicate information needed for operations.

2. Class Diagram



In this Class Diagram, we summarize the definitions of each class with expected attributes and methods. The relationship between each class is also depicted in the diagram. The HBnB system has been broken down into 4 classes which handles the functionalities described in the business layer logic above. The four classes include:

User:

The User class represents the description of the users interacting with the presentation layer of the HBnB system.

Place:

The Place class represents the description of the properties listed by the host on the HBnB system.

Review:

The Review class represents the reviews and ratings of different properties instantiated from Place by User on the HBnB system.

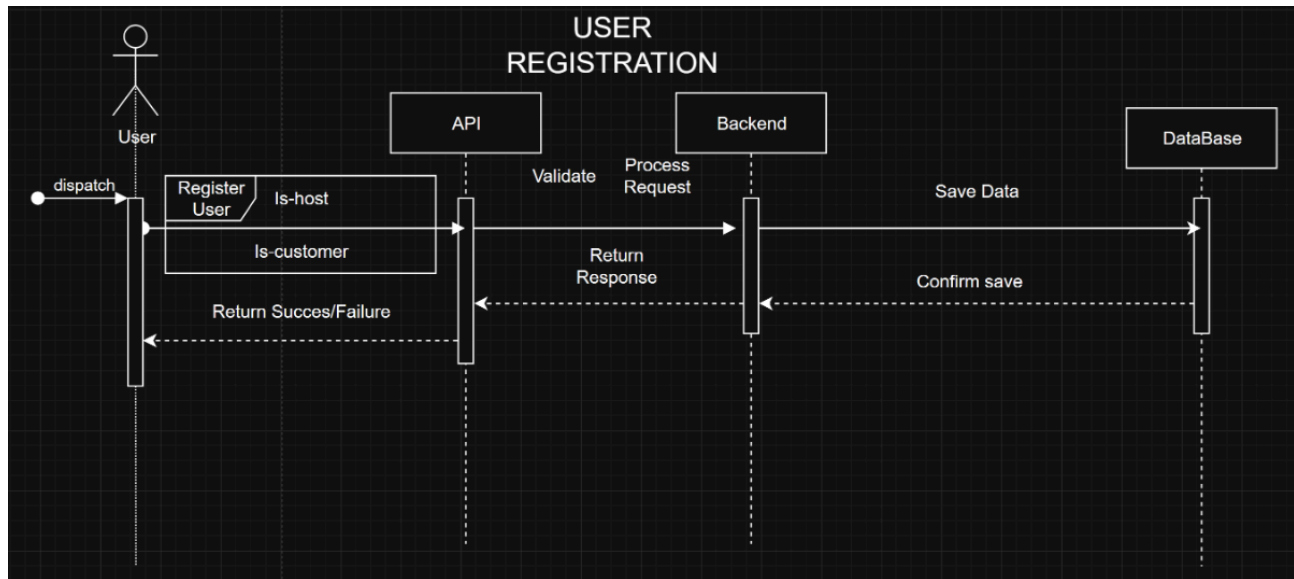
Amenities:

The Amenities class represents the room description of the properties instantiated by Place.

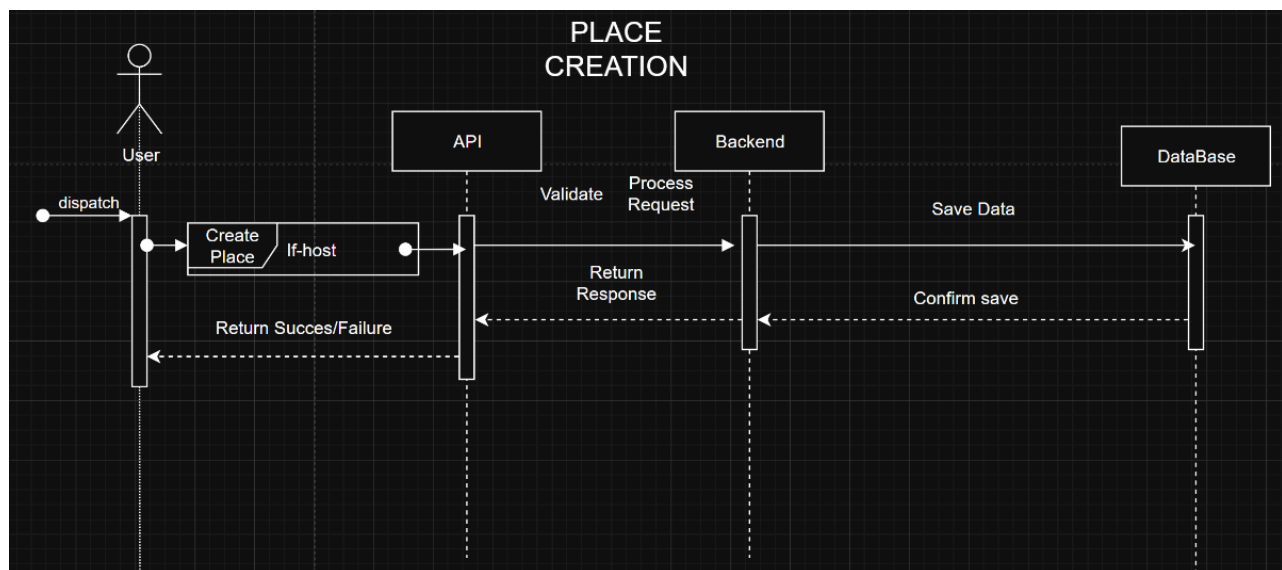
3.Sequence Diagram

The sequence diagrams is showing how the API is connecting the user and the backend while also demonstrating how the backend is handling database interaction.

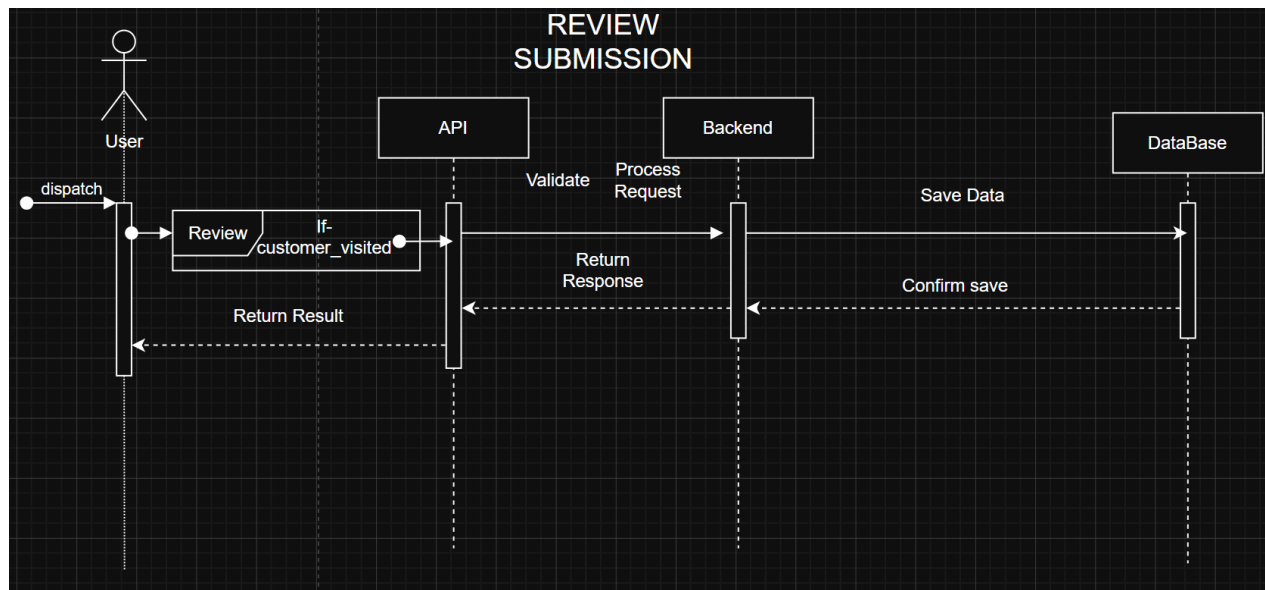
Here is a list of diagrams now!



User will commence in creating an account through which API will validate the information, checking for errors and if successful will process the information through the methods, saving the users information to the Database.
Returns confirmed information was saved. if saved,
Backend returns response to API.
API returns the success or failure to user if their account has been created.

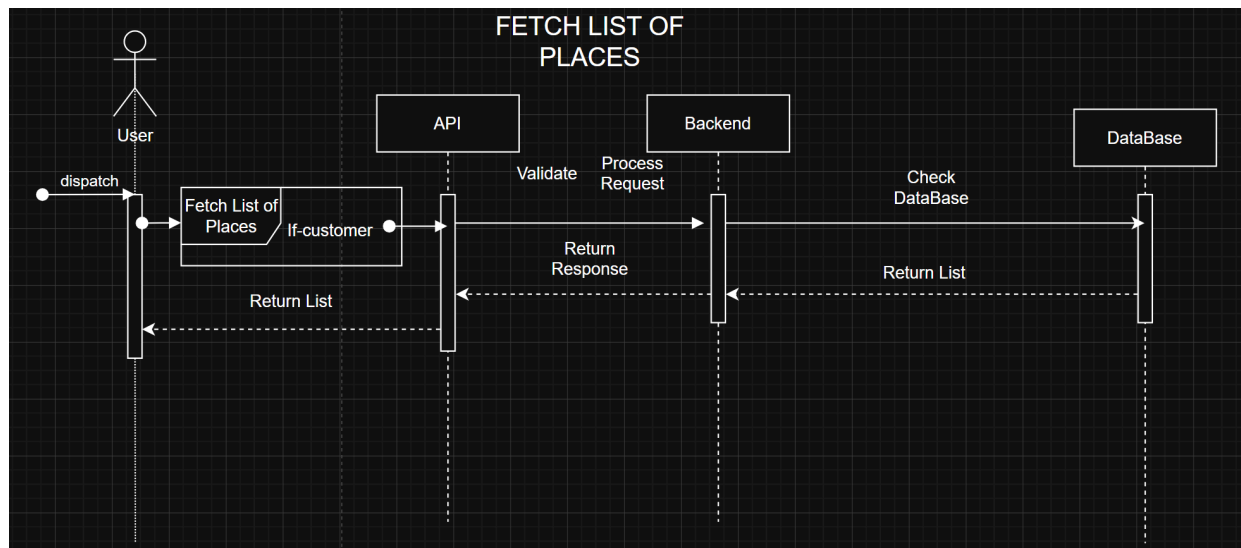


If user is a host, they can Create a listing of their place, the information will be validated(Checked for errors) than the request will be processed through methods, than passed to database. Data will be saved, confirmed if saved and backend will return response to API. API will return a success if data was saved for the property becoming valid.



The purpose of this diagram is to describe the review submission process.

Customers can only review places when they have used them. So, before the customer is allowed to save their review, a validation will check if they have used the place whose result will be passed on by the API to the backend. The backend receives the result and if true will process the request which will save the data to the database. The database will confirm the save and returns that result to the user. This review will be displayed on the place page.



The purpose of this diagram is to describe the process of fetching a list of places based on user criteria (might be location, cost, availability).

A user will initiate a search of places in the website. This request will be passed on by the API to the backend where the request is first validated (spelling check, user status(if customer), specifications).

After validation, the backend processes the request which is then passed on to the database where a search is initiated and the resulting list is returned to the backend. The backend will inform the API of the success/failure of the search which upon success the list is passed to the user.